Week 6 Progress Report

ECE501: Digital Image Processing | GROUP 19 | October 25, 2026

Project Topic: Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

Deliverable: LBP variation study (V1 to V4), GLCM variations, LBP + GLCM fusion, and an honest evaluation of where the current detectors fail

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Objectives of the Week

- Implement the LBP variations proposed in Week 5 (illumination normalisation, multiscale LBP, post-processing) and measure each one separately.
- Implement the proposed GLCM variations.
- Implement LBP + GLCM fusion (score-level, AND, OR) and compare it fairly against the individual methods.
- Evaluate with ground-truth masks using image-level and pixel-level metrics, including IoU and Dice.

Gabor features, rotation tests and the published-AITEX comparison were planned for Week 6 but have moved to Week 7, because this week went into the LBP/GLCM variations, fusion, and diagnosing the results.

Work Done This Week

A. Dataset and evaluation setup

- Dataset: AITEX Fabric Image Database (AFID) [3], fabric type 03, with pixel-level ground-truth masks.
- All images are converted to grayscale and resized to 256 x 256 (masks use nearest-neighbour resizing and are binarised).
- The defect-free images are split 70/30 by image (fixed seed): 70% build the normal reference, 30% are held out only to calibrate the threshold.
- Test set: 10 images in total. From the confusion counts, 5 are defective (with masks) and 5 are separate defect-free images. Normal images get an all-zero mask.

- Patches are 16 x 16 with step 8. Each patch gets a chi-square distance between its LBP histogram and the reference normal histogram. Overlapping patch scores are averaged into a pixel-level anomaly map.
- The pixel threshold is the 99th percentile of the anomaly scores on the held-out normal images. An image is called defective if at least 12 pixels are flagged.
- Metrics: image-level accuracy, precision, recall and F1; pixel-level accuracy, precision, recall, F1, IoU and Dice; plus TP/TN/FP/FN counts.

B. LBP variations (V1 to V4)

Each variation adds one change to the previous one, so the effect of each change can be read from the table.

Variation	What changes	Problem it targets
V1	Basic uniform LBP (P=8, R=1), no CLAHE	Baseline
V2	V1 + CLAHE (clip 2.0, 8 x 8 tiles)	Brightness mismatch between images
V3	V2 + three scales (8,1), (16,2), (24,3), score maps averaged	Small and larger texture changes
V4	V3 + median filter, morphological closing, removal of regions under 12 px	Scattered tiny false regions

Table 1: Image-level results (10 test images)

Variation	Accuracy	Precision	Recall	F1
V1	0.40	0.4444	0.80	0.5714
V2	0.40	0.4444	0.80	0.5714
V3	0.40	0.4444	0.80	0.5714
V4	0.40	0.4444	0.80	0.5714

These numbers correspond to 4 true positives, 1 false negative, 5 false positives and 0 true negatives for every variation: **all 5 defect-free test images were flagged as defective**, and 4 of the 5 defective images were caught.

Table 2: Pixel-level results

Variation	Pixel Acc.	Pixel Recall	TP	FP	FN	FP as % of normal pixels	Threshold
V1	0.7414	0.0353	3	169,405	82	25.9%	0.0943
V2	0.7483	0.0353	3	164,861	82	25.2%	n/a

Variation	Pixel Acc.	Pixel Recall	TP	FP	FN	FP as % of normal pixels	Threshold
V3	0.7800	0.0353	3	144,125	82	22.0%	0.1305
V4	0.7800	0.0353	3	144,111	82	22.0%	0.1305

Pixel precision is about 0.00002 and IoU and Dice are effectively 0 for all four variations (the code table shows them rounded to 0.0).

Observations

- **CLAHE (V1 to V2)** reduced false-positive pixels only slightly (169,405 to 164,861, about 2.7%).
- **Multiscale LBP (V2 to V3)** gave the largest improvement: false-positive pixels fell by about 12.6% relative to V2 (about 15% relative to V1), and pixel accuracy rose from 0.7414 to 0.7800.
- **Post-processing (V3 to V4)** removed only 14 pixels. The flagged regions come from 16 x 16 patches, so they are already larger than the 12-pixel minimum area, and the cleanup has almost nothing to remove. For this patch-based score map, V4 is effectively equal to V3.
- **Defect pixels are barely found.** Only 3 of 85 ground-truth defect pixels are detected in every variation. The ground-truth defects at 256 x 256 are tiny (85 pixels over the 5 defective images, roughly 17 pixels per image on average), while one 16 x 16 patch covers 256 pixels. A single flagged patch is therefore already much larger than the whole defect, which limits the IoU and precision this window size can reach.
- **Image-level metrics are identical across variations.** Because one flagged patch (at least 12 pixels) makes an image "defective", every normal image is flagged regardless of the variation. The image-level decision rule is too lenient to separate the variations.
- **The recurring false alarm is the edge band.** In the example outputs (e.g. 0090_002_01 (4).png, V3, and 0056_019_06 (1).png, V1) the strongest anomaly scores sit on the bright, differently textured strip along the left edge of the fabric, where the reference histogram does not apply. This is the same edge-band problem listed in Week 5 and it is not fixed by CLAHE or multiscale LBP.
- **Reading the overlay figures:** the overlay uses the matplotlib colormap `autumn` with values 0 and 1, which draws background (0) in red and flagged pixels (1) in yellow. Yellow therefore marks the flagged region and the large red area is the unflagged region. The metrics are not affected, but the colours are misleading. This will be changed to a single-colour overlay in Week 7.

C. GLCM variations

GLCM features follow Week 5 (contrast, dissimilarity, homogeneity, energy, correlation, averaged over four angles). Variations implemented this week:

- Three variants were compared: Baseline (the Week 5 GLCM detector), Enhanced_KNN (enhanced features, scored by the mean distance to the k nearest normal patches) and Enhanced_Mahalanobis (the same enhanced features, scored by Mahalanobis distance). [State exactly which feature enhancements were added.]

Results on one wide fabric strip. No ground-truth mask was used here, so these are detected-area figures and visual checks, not pixel metrics:

Variant	Prediction	Detected area (%)
Baseline	Possible defect	4.93
Enhanced_KNN	Possible defect	4.86
Enhanced_Mahalanobis	Possible defect	4.87

- All three variants report a possible defect and mark a compact region near the centre of the strip, close to the visible mark in the original image. This is much more localised than the LBP results.
- In the fusion run, GLCM training took 19.2 s and testing 1.1 s, against LBP training 1.6 s and testing 2.3 s. GLCM is about 12 times slower to train but faster to test.
- The three variants differ by only 0.07 percentage points in detected area (4.86% to 4.93%), so on this image the k-nearest and Mahalanobis scoring did not change the result measurably. The 71.61% GLCM figure in Section D comes from a different image and setup (32 x 32 window, step 16), so the two are not directly comparable. IoU and Dice for GLCM need ground-truth masks and are planned for Week 7.

D. LBP + GLCM fusion

Fusion was compared on a wide fabric strip (lighting-flattened) using a 32 x 32 window with step 16 and 8 training images. Five detectors were compared: LBP, GLCM, score-level FUSION, AND (LBP & GLCM) and OR (LBP | GLCM).

Table 3: Fusion comparison on one test image

Method	Flagged area %	Prediction
LBP	83.03	Defective
GLCM	71.61	Defective
FUSION	80.77	Defective
AND (LBP & GLCM)	68.97	Defective
OR (LBP | GLCM)	85.67	Defective

Observations

- Every method flags 69% to 86% of the image. The visible defects in this strip are thin: a bright vertical band near the left and a thin vertical line near the right. None of the detectors isolates them. A result that flags most of the image is not localisation.
- The anomaly score maps are highest in the centre of the strip and lowest toward the left and right ends. The score does not follow the defects. This points to a position- or lighting-dependent change in the normal texture and a threshold set too low, not to defect sensitivity. This is our interpretation and has not yet been tested.

- AND is the most conservative combination (68.97%) and OR the most permissive (85.67%), as expected. The score-level fusion (80.77%) lies between LBP and GLCM and gives no clear gain over either at the current margins.
- The program output itself notes that a method showing too much red needs a higher MARGIN. No margin tuning was done this week, so these figures are before tuning.
- This comparison is a single image, so it shows behaviour, not performance. It is not a metric on the test set.

E. What this week shows

1. Among the LBP variations, only multiscale LBP gave a measurable improvement; CLAHE helped slightly and post-processing had no effect.
2. The LBP variations and the fusion rules over-flag; the GLCM variants flagged only about 5% of the one strip tested. About 22% to 26% of normal pixels are flagged at the pixel level, and all defect-free test images are flagged at the image level.
3. The thresholds do not carry over from the held-out normal images to the test images. With only 8 to 10 normal images in total, the calibration set is very small (only a few images after the 70/30 split), so the 99th percentile is not a stable estimate. The fabric images also differ in overall brightness (compare the dark image in Fig. 4 with the bright image in Fig. 7), which a mean reference histogram does not handle.
4. Window size and defect size are mismatched: the defects are smaller than one window.
5. The edge band is the main source of false alarms and needs its own handling.

Changes from Week 5

Week 5	Week 6
Two-class LBP and GLCM implemented	LBP variations V1 to V4, GLCM variations and fusion implemented and evaluated
Held-out numbers not yet available	First held-out numbers, with IoU and Dice, on 10 test images
Variations only proposed	Illumination normalisation, multiscale and post-processing tested for LBP
Gabor planned for Week 6	Moved to Week 7

Note: the V1 to V4 study uses a normal-only reference with a chi-square distance and a percentile threshold, so that each change can be isolated. It is not the two-class log-likelihood-ratio model from Week 5, and the two have not yet been compared on the same split.

Plan for Next Week (Week 7)

- Fix the threshold: use more held-out normal images, a per-image or per-region normalisation, and tune the fusion MARGIN on validation images instead of fixing it.
- Handle the edge band: mask it out or give it a separate reference.
- Use smaller and thin windows (8 x 8, 8 x 32, 32 x 8) because the defects are smaller than 16 x 16.

- Change the image-level rule from "at least 12 pixels" to a minimum number of flagged patches or a score-based decision, and report AUC so the result does not depend on one threshold.
- Change the overlay colours so flagged pixels are clearly shown.
- Run the two-class model on the same split so the one-class and two-class results can be compared.
- Add Gabor features, cTPR, the patch-size sweep, the rotation test and the published AITEX comparison (as planned in Week 5).
- Complete the metric tables with cross-validation over images, run times and per-defect-type recall.

Conclusion

This week LBP and GLCM were extended with variations and fused. Multiscale LBP gave the only clear improvement, and CLAHE and post-processing added little. The LBP variations and all fusion rules still flag a large part of the fabric, so localisation of the real defects is not yet achieved: on the held-out test set only 3 of 85 defect pixels are found, IoU is near zero, and every defect-free test image is flagged. The main causes appear to be a small and unrepresentative calibration set, the edge band, brightness differences between images, and windows larger than the defects. Week 7 will address these before adding Gabor features.

References

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